

Spatial-temporal Prediction of Algal Bloom

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Abstract—We present an application of spatial-temporal prediction to track algal blooms. Algal bloom is an important water quality events in marine, coastal and estuarine environments. For a day, we first identify an area with anomalous algal growth represented by spatial points in the gridded data where values of Chlorophyll-a (indicator for algal bloom) are above a threshold chosen by domain scientists. To represent the shape of the algal bloom area, we create convex hull from spatial gridded points. We then find the radii from centroid of the convex hull. The radii are further used as features in predicting spatial region of the algal bloom using regression techniques. We also predict the centroid (represented in latitude and longitude) of an algal bloom area to track whether bloom area is moving. Experimental results show that our approach can reasonably predict algal bloom area and its centroid one day ahead using features from previous day. The prediction technique benefits towards decision support systems for aquaculture industry and environmental departments.

I. INTRODUCTION

Spatial and temporal monitoring and tracking of environmental events such as cyclones, hurricanes [1], pollution(e.g. oil-spills [2], [3], [4], ammonia-spills [5]) are important in assessing and managing their effects. Algal bloom is an event that occurs in marine, coastal and estuarine water when algae grows excessively with suitable environmental conditions. Algal bloom covers a region in the water and may last for some days. The area affected by an algae bloom can have effects on fish kills and further impacts on human health when fish exposed by toxic algae are consumed [6]. The aquaculture industry needs to know which area in the water is affected by the algal bloom so that they can take management and operational decisions. Besides industry needs, the environmental authorities provide decisions and advice to public for recreational activities in the algal bloom affected water. Thus, monitoring and tracking of algal blooms [7], [8], [9], [10] in marine, coastal and estuarine areas is becoming necessary because of their negative impacts on aquaculture industries and public health. Also, tracking and predicting algal bloom events can be used in mission planning [11] using mobile sensing platforms in sampling water quality variables when an algal bloom occurs.

In this paper, we study an approach in predicting and tracking algal blooms using biogeochemical (BGC) model data for Derwent Estuary, located at the south east of Tasmania, Australia.

Our contribution to this paper is described below.

- In spatial and temporal prediction, we consider the application of convex hull for the spatial points in the grid

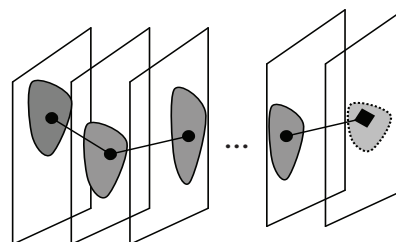


Fig. 1. Sketch of spatial temporal prediction of algal bloom

where chlorophyll-a is above the threshold for a day. The convex hull represents the shape of the algal bloom area. We further extract radii from the centroid as features of the convex hull to represent the approximate shape of the algal bloom. This process is performed for some consecutive days when the chlorophyll-a is higher than the threshold for an area. The radii features are used in building a prediction model for algal bloom using machine learning techniques.

- We also build a prediction model for the centroid of the algal bloom for tracking the location of the algal bloom. The sketch of the problem is shown in figure 1. In the figure, shaded region is the area and shape where chlorophyll-a is higher than the threshold. Centroid is represented using dark dots and predicted centroid is represented as diamond. Predicted shape is represented in dotted line.

This paper is organized as follows. The methods of feature extraction are explained in Section II. We present the prediction model of algal bloom shapes and their results in Section III. The prediction model for the algal bloom centroid and associated results are presented in Section IV. Finally we conclude in Section V.

II. METHODS

In this research, we have used biogeochemical(BGC) model data from April 2003 to March 2004 for the Derwent estuary in South East of Tasmania, Australia. The model consists of 212x52 surface grid cells [12]. The BGC model has 23 vertical layers with 0.5 meter resolution to 8 meter depth and we have considered the values at surface only.

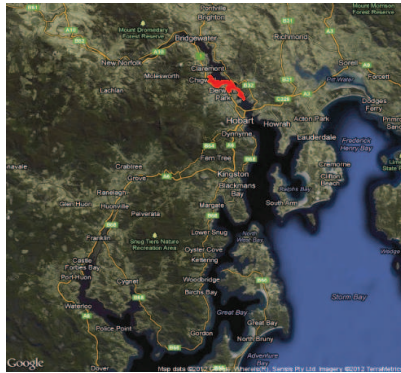
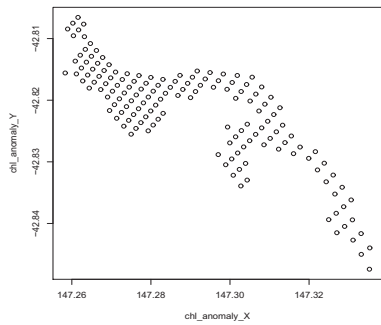


Fig. 2. (a)Chl-a anomaly points in the grid on 12/23/2003 (b)Chl-a anomaly on 12/23/2003 plotted using Google Maps

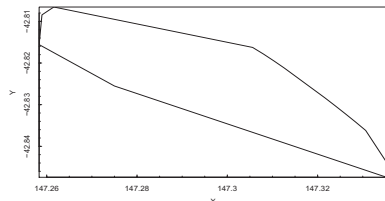


Fig. 3. Convex hull calculated from Chl-a anomaly points on 12/23/2003

A. Representing and Extracting Shape Features of an Algal Bloom

In representing shape of an algal bloom, we use the following steps.

*Step 1(points of the anomaly):*We extract the Chl-a anomaly of a day from the surface grid where the Chl-a values of the grid cells (i.e. points) are higher than a threshold value defined by a domain expert. Further we extract consecutive daily patches where the values of the grid points are higher

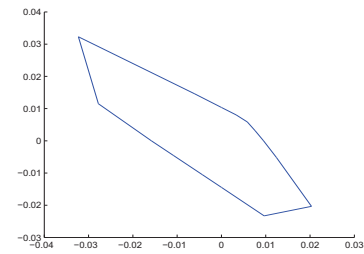


Fig. 4. Shape represented by radius from centroid of the convex hull on 12/23/2003

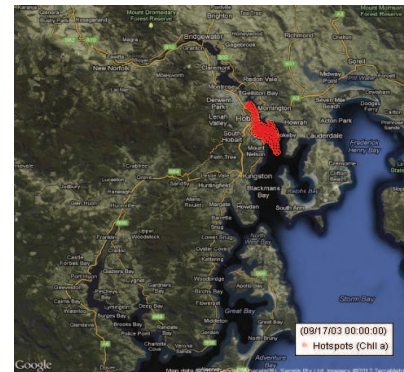
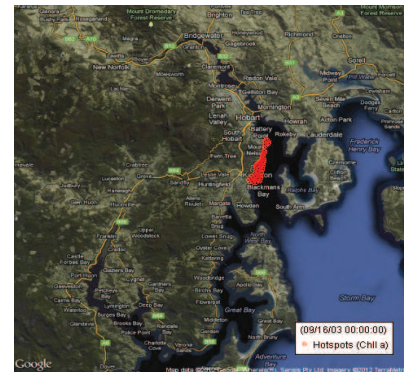


Fig. 5. Chl-a anomaly on 09/16/03 and 09/17/03 together considered as a patch

than the threshold. As an example, we show the region where there is a Chl-a anomaly on 12/23/2003 plotted in Fig.2(a) and in Fig.2(b) using Google Maps [13].

Step 2(shape of the anomaly points): We calculate the convex hull of each Chl-a anomaly that is defined by the grid points. This convex hull gives the spatial shape (i.e. contour) of the Chl-a anomaly. We show the convex hull of the anomaly on 12/23/2003 in Fig.3.

Step 3(radius of the convex hull): For each of the convex hulls, we now calculate the radius from the centroid of the convex hull. From the centroid, we consider 16 radii separated at 22.5° angles from the centroid. These 16 radii represent the

TABLE I
PATCH 1

Radius	Actual	SVR	RBF	LR
c1	0.01	0.015	0.025	0.01
c2	0.008	0.012	0.014	0.013
c3	0.016	0.02	0.019	0.019
c4	0.009	0.011	0.021	0.016
c5	0.023	0.025	0.035	0.012
c6	0.007	0.011	0.013	0.011
c7	0.017	0.013	0.02	0.01
c8	0.007	0.012	0.022	0.016
c9	0.023	0.028	0.031	0.027
c10	0.008	0.012	0.013	0.012
c11	0.018	0.011	0.02	0.022
c12	0.007	0.011	0.02	0.014
c13	0.011	0.016	0.018	0.016
c14	0.01	0.015	0.016	0.015
c15	0.014	0.012	0.021	0.006
c16	0.007	0.012	0.021	0.016

approximate shape of the previously computed convex hull shape. We consider 16 radii to represent the shape in this paper but we can increase more radii for better representation of the shape. We show the shape of the anomaly on 12/23/2003 using radii calculated from the convex hull in Fig.4.

III. ONE DAY AHEAD PREDICTION OF THE SHAPE OF AN ALGAL BLOOM

In this section, we present the technique and results of prediction of the shape of an algal bloom one day ahead. We consider the following mapping function in predicting the shape of an algal bloom one day ahead:

$y_{i,j} = f(x_{i-1,j})$ where x, y mean radius of an algal bloom, the index i is the current day, the index $i-1$ is the previous data and the index $j = [1, \dots, 16]$. We consider f as a regression function where the radius of the algal bloom on the previous day are the inputs. In the regression function, we model each of the radii of the current shape of Chl-a as the output from the regression function f .

We now present the results in predicting the shape of an algal bloom from different patches. By patch, we mean the Chl-a anomaly events that happen over consecutive days. An example of a patch is shown in Fig.5.

A. Results

In data preparation for the learning models shown in the previous section, we used both R [14] and MATLAB [15] programming languages. We use data mining and machine learning tool WEKA [16] for implementing models using support vector regression (SVR), radial basis functions (RBF) and linear regression (LR). In the experiments, we consider four Chl-a anomaly patches. We only show predictions using SVR and RBF in the figures.

[Patch 1] Chl-a anomaly from 12/23/2003 to 12/30/2003: The actual and predicted values of the radii for the anomaly on 12/30/2003 are given in Table I. We show the actual shape of the Chl-a anomaly on 12/30/2003 in Fig.6(a). For patch 1, SVR performs better than RBF and LR.

TABLE II
PATCH 2

Radius	Actual	SVR	RBF	LR
c1	0.037	0.025	0.025	0.025
c2	0.012	0.023	0.022	0.023
c3	0.02	0.038	0.034	0.039
c4	0.037	0.021	0.023	0.022
c5	0.057	0.05	0.049	0.048
c6	0.01	0.02	0.019	0.019
c7	0.033	0.032	0.04	0.044
c8	0.028	0.017	0.018	0.018
c9	0.039	0.07	0.066	0.071
c10	0.011	0.02	0.02	0.02
c11	0.038	0.018	0.045	0.012
c12	0.026	0.018	0.045	0.012
c13	0.017	0.034	0.032	0.03
c14	0.013	0.025	0.024	0.023
c15	0.039	0.027	0.032	0.027
c16	0.028	0.018	0.019	0.019

TABLE III
PATCH 3

Radius	Actual	SVR	RBF	LR
c1	0.015	0.024	0.03	0.02
c2	0.011	0.013	0.015	0.011
c3	0.016	0.02	0.024	0.015
c4	0.015	0.019	0.025	0.018
c5	0.026	0.039	0.044	0.031
c6	0.009	0.011	0.013	0.009
c7	0.041	0.042	0.051	0.03
c8	0.012	0.016	0.02	0.014
c9	0.036	0.033	0.04	0.032
c10	0.009	0.011	0.013	0.009
c11	0.047	0.051	0.062	0.037
c12	0.011	0.016	0.019	0.014
c13	0.016	0.018	0.023	0.016
c14	0.011	0.013	0.016	0.01
c15	0.023	0.031	0.039	0.029
c16	0.011	0.018	0.022	0.015

We show the predicted shape using support vector regression in Fig.6(b) and using radial basis functions (RBF) in Fig.6(c).

[Patch 2] Chl-a anomaly from 12/11/03 to 12/16/03: The actual and predicted values for the anomaly on 12/16/2003 are given in Table II. We show the actual shape of the Chl-a anomaly on 12/16/2003 in Fig.7(a). We show the predicted shape using support vector regression in Fig.7(b) and using RBF in Fig.7(c). In the case of patch 2, RBF performs better than SVR and LR.

[Patch 3] Chl-a anomaly from 11/27/03 to 12/02/03: The actual and predicted radii values for the anomaly on 12/02/2003 are given in Table III. The actual shape of the Chl-a anomaly on 12/02/03 is shown in Fig.8(a). The predicted shape using support vector regression is shown in Fig.8(b). The predicted shape using RBF is shown in Fig.8(c). LR shows better accuracy than RBF and SVR.

[Patch 4] Chl-a anomaly from 09/21/03 to 09/26/03: The actual and predicted radii values for the anomaly on 09/26/2003 are given in Table IV. The actual shape of Chl-a anomaly on 09/26/2003 is shown in Fig.9(a). The predicted shape on 09/26/03 using SVR is shown in Fig.9(b) and using RBF in

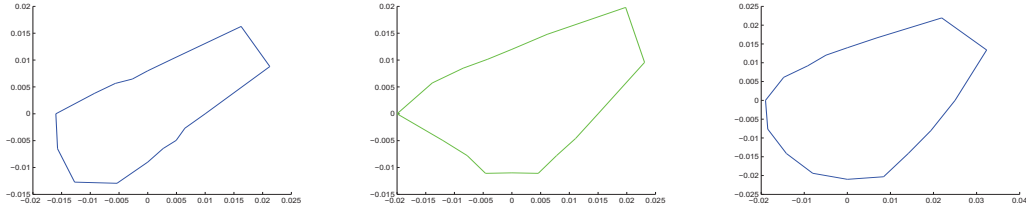


Fig. 6. (a)Actual shape of the Chl-a anomaly, (b)Predicted shape using SVR, (c)Predicted shape using RBF on 12/30/03

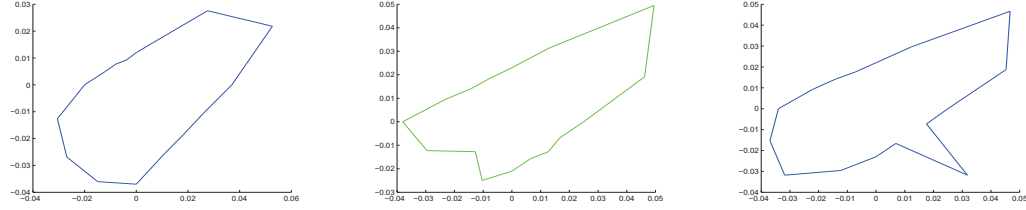


Fig. 7. (a)Actual shape of the Chl-a anomaly, (b)Predicted shape using SVR, (c)Predicted shape using RBF on 12/16/03

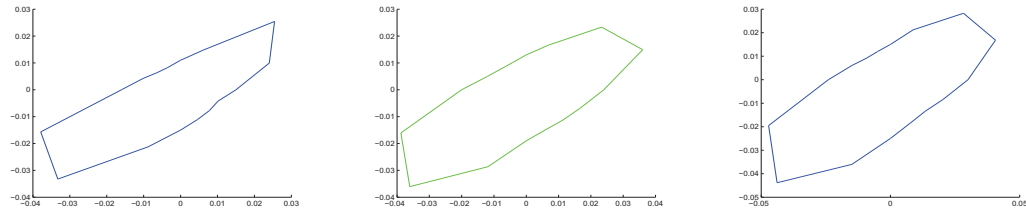


Fig. 8. (a)Actual shape of the Chl-a anomaly, (b)Predicted shape using SVR, (c)Predicted shape using RBF on 12/02/03

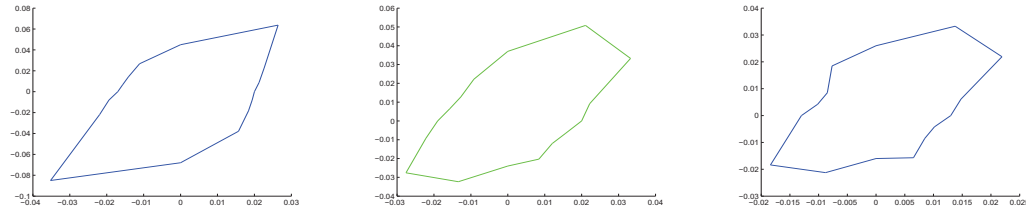


Fig. 9. (a)Actual shape of the Chl-a anomaly, (b)Predicted shape using SVR, (c)Predicted shape using RBF on 09/26/03

Fig.9(c). In this case, SVR performs better than RBF and LR.

We present the mean absolute percentage error (MAPE) in table V. Mean absolute percentage error(MAPE):

$$MAPE = 100 \frac{1}{n} \sum_{i=1}^n \frac{|m_i - p_i|}{m_i} \quad (1)$$

where m is the actual value and p is the predicted value.

We consider the MAPE for each algal bloom patch events and the average for all patch events. We find that support vector regression performs best and RBF performs worst among the three regression techniques in terms of average accuracy.

IV. ONE DAY AHEAD PREDICTION OF THE CENTROID OF THE SHAPE OF AN ALGAL BLOOM

In this Section, we present one day ahead prediction of the centroid of the shape of an algal bloom. The need for the prediction of the centroid of the shape of an algal bloom is to track the location of the shape. We use the function $x_i, y_i = f(x_{i-1}, y_{i-1}, cv_{i-1}, cd_{i-1})$ where i is the index of the current day, $i-1$ is the index of the previous day, x is the longitude of the centroid, y is the latitude of the centroid, cv is the current velocity at the centroid and cd is the current direction at the centroid.

In predicting the centroid of the shape of Chl-a anomaly, we not only consider centroid of the previous day. We also

TABLE IV
PATCH 4

Radius	Actual	SVR	RBF	LR
c1	0.02	0.02	0.013	0.016
c2	0.045	0.037	0.026	0.024
c3	0.017	0.019	0.013	0.013
c4	0.068	0.024	0.016	0.018
c5	0.023	0.024	0.016	0.018
c6	0.029	0.024	0.02	0.018
c7	0.021	0.024	0.016	0.017
c8	0.041	0.022	0.017	0.017
c9	0.032	0.047	0.031	0.031
c10	0.02	0.018	0.012	0.013
c11	0.031	0.039	0.026	0.026
c12	0.026	0.017	0.012	0.014
c13	0.069	0.055	0.036	0.041
c14	0.017	0.017	0.011	0.012
c15	0.092	0.035	0.023	0.025
c16	0.021	0.017	0.011	0.014

TABLE V

MEAN ABSOLUTE PERCENTAGE ERROR(MAPE) IN SHAPE PREDICTION

	Patch 1	Patch 2	Patch 3	Patch 4	Average
SVR	41.06	57.22	28.22	24.69	37.80
RBF	88.42	53.13	54.80	39.99	59.08
LR	56.53	58.56	15.84	36.1	41.76

consider the current velocity and the current direction as inputs. The reason for choosing current velocity and direction is that these variables have a physical effect in the movement of an algal bloom patch.

A. Results

We show the one day ahead prediction of the shape of Chl-a anomaly patch from 12/11/03 to 12/17/2003 in Fig.10. We use training data consisting of the centroid, current velocity and direction from 12/11/03 to 12/16/03 using support vector regression while the testing on data on 12/17/03. In case of predicting the centroid of the shape of an algal bloom for patch 1, SVR performed better than RBF and LR.

V. CONCLUSIONS AND FUTURE WORK

In this paper, we have presented a technique for predicting the shape of a Chl-a anomaly that covers a geographic area. We then showed the results of one day ahead prediction of Chl-a anomaly shape using features of the previous days shape using machine learning techniques. The prediction of the centroid of the shape for a patch is also shown. Our results show reasonable accuracy in predicting shape and centroid of the shape of an algal bloom. In future work, we plan to use more radii from the centroid of the shape for better accuracy in prediction. We also plan to investigate the splitting and merging of algal blooms as affected by environmental variables.

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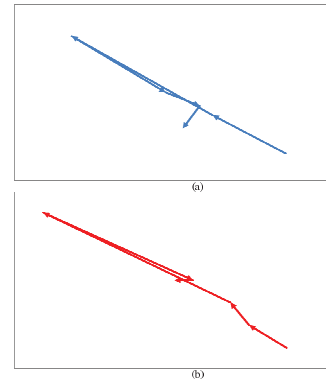


Fig. 10. (a) Actual trajectory of the centroid of algal blooms, (b) Predicted trajectory using SVR

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